An Intelligent Assistant for Fertility Prediction, Wellness, and Support

25-26J-246

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

An Intelligent Assistant for Fertility Prediction, Wellness, and Support (IVF Success Prediction)

25-26J-246

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

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DECLARATION

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DECLARATION

I hereby declare that the work presented in this report of proposal is my own and has not been submitted elsewhere. All sources of information have been duly acknowledged. This report is submitted in partial fulfillment of the requirements of the above degree program, and it reflects the research and development plan for my individual component of the project.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

Predicting the success of in vitro fertilization (IVF) for Sri Lankan couples is critical for reducing uncertainty, guiding clinical decisions, and minimizing emotional and financial burdens. Current IVF prediction models, often developed for Western or Chinese populations, lack dynamic adaptability, local calibration, and integration of lifestyle factors relevant to Sri Lanka. This study reviews existing AI and machine learning (ML) methods for IVF success prediction, focusing on dynamic multi-landmark models, explainable nonlinear frameworks, and center-specific calibration strategies. Drawing from transferable components in multi-stage prediction, cumulative live birth estimation, and localized recalibration, we propose a novel integration strategy. This combines fine-tuned logistic regression and ensemble ML models (e.g., Random Forest, XGBoost) with restricted cubic splines for nonlinear effects, incorporating demographic, lifestyle, medical history, hormonal markers, and cycle-specific clinical parameters. The approach includes continuous learning pipelines to adapt to new outcomes and center-specific variations in Sri Lanka. Key gaps addressed include Sri Lanka-specific validation, leakage control in multi-landmark predictions, and handling of selection bias and missing data. The proposed system aims to deliver personalized, explainable, and locally calibrated IVF success probabilities, enhancing decision-making for Sri Lankan couples and clinicians. Challenges include the absence of local annotated datasets, regional variations in clinical practices, and the need for robust drift monitoring.

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Table of Contents

Contents

Abstract ii

Table of Contents iii

List of Figures v

List of Tables vi

List of Abbreviations vii

1.0 INTRODUCTION 8

1.1 Background and Literature 8

1.2 Research Gap 11

1.3 Research Problem 14

2.0 Objectives 17

2.1 Main Objective 17

2.2 Specific Objectives 17

3.0 METHODOLOGY 19

3.1 Requirement Gathering 19

3.1.1 Past Research Analysis 19

3.2 Feasibility Study 20

3.2.1 Technical Feasibility 20

3.2.1.1 Knowledge of Technologies 20

3.2.1.2Knowledge on Tools 21

3.3 System Analysis 22

3.3.1 Software Solution Approach 22

3.3.2 Tools & Technology 25

3.4 Project Requirements 27

3.5 Testing 29

3.6 Timeline 30

3.7 Risk Management Plan 31

3.8 Communication Management Plan 31

3.8.1 Communication Objectives: 32

3.8.2 Communication Media: 32

4.0 COMMERCIALIZATION 34

5.0 BUDGET 35

6.0 SUMMARY 36
References 37

List of Figures

Figure 1- IVF Success Prediction System Architecture 10
Figure 2- Feature Importance in IVF Success Prediction 13
Figure 3: System Architectural diagram 24
Figure 4: Work breakdown Structure 26
Figure 5:Gantt Chart for Project Schedule Management 30

List of Tables

Table 1-: Risk Management Plan 31
Table 2- Communication Management Plan 32
Table 3-Cost management plan 35

List of Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

DL – Deep Learning

IVF – In Vitro Fertilization

ART – Assisted Reproductive Technology

AMH – Anti-Müllerian Hormone

FSH – Follicle-Stimulating Hormone

LH – Luteinizing Hormone

ROC-AUC – Receiver Operating Characteristic – Area Under the Curve

API – Application Programming Interface

SaaS – Software as a Service

DB – Database

UI/UX – User Interface / User Experience

AWS – Amazon Web Services

PWA – Progressive Web Application

1.0 INTRODUCTION

1.1 Background and Literature

The growing demand for in vitro fertilization (IVF) in Sri Lanka, coupled with limited access to advanced fertility treatments and high emotional and financial costs, underscores the need for accurate, personalized IVF success prediction tools [7], [20]. Existing prediction models, primarily developed for Western or Chinese populations, often rely on static, generalized approaches that fail to account for Sri Lanka's unique demographic, lifestyle, and clinical variations [4], [6]. This has driven the development of AI and machine learning (ML)-based methods to provide dynamic, explainable, and locally calibrated predictions of IVF success, enabling better decision-making for couples and clinicians [1], [3], [18].

A key component of advanced IVF prediction is dynamic multi-landmark modeling, which updates success probabilities at critical stages of the IVF cycle (e.g., consultation, post-stimulation, post-lab, pre-transfer). Studies such as Grzegorczyk-Martin et al. have demonstrated the feasibility of multi-stage models that integrate diverse features like demographics, hormonal markers, and cycle-specific parameters to refine predictions as new data accrues [1]. Similarly, Xia et al. developed phase-specific models for cumulative live birth rate (CLBR), using logistic regression with restricted cubic splines to capture nonlinear effects of age and antral follicle count (AFC) [2]. These approaches highlight the potential for dynamic prediction frameworks adaptable to Sri Lanka's clinical settings.

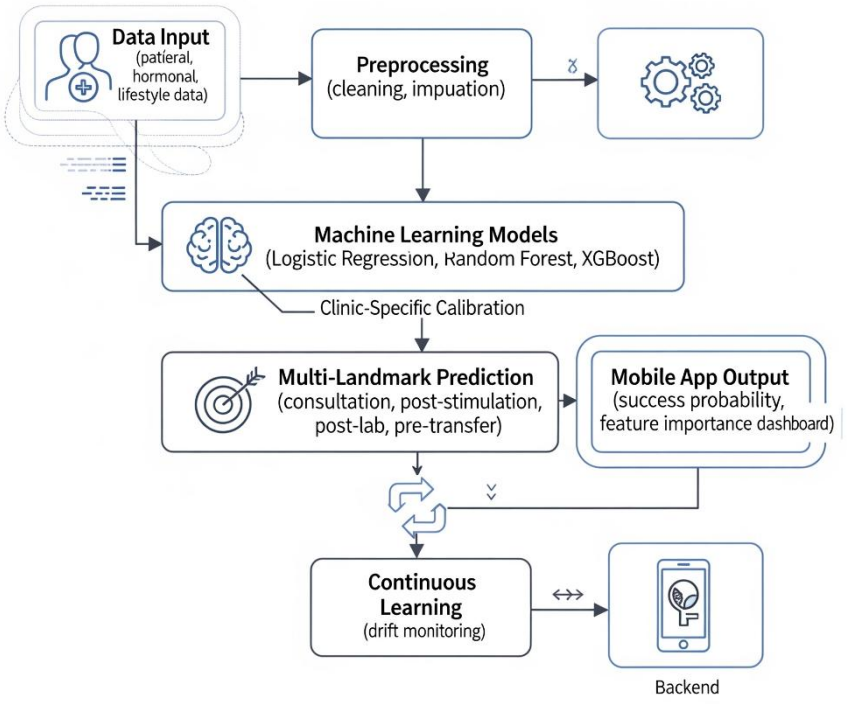
Another critical aspect is explainable modeling, which ensures predictions are interpretable for clinicians and patients. The use of restricted cubic splines in Xia et al. provides a flexible, clinically interpretable approach to modeling nonlinear relationships, such as the impact of age or AMH levels on IVF outcomes [2]. Ensemble methods like Random Forest and XGBoost, as explored in Nguyen et al. and Cai et al., offer feature importance scores, identifying key predictors like BMI or egg retrieval numbers [3], [4]. These methods are promising for Sri Lanka, where transparent predictions can build trust and support shared decision-making [18].

Cumulative live birth prediction over multiple IVF cycles is a well-established estimand in the literature, particularly in the McLernon family of studies. These models predict CLBR over up to six complete cycles, incorporating pretreatment and post-treatment factors such as age, infertility duration, and egg yield, with robust internal and external validation [5], [6], [7], [8], [9]. Leijdekkers et al. validated these models in a Dutch cohort, showing that minor recalibration restores performance across jurisdictions, a critical insight for adapting models to Sri Lanka [9]. Center-specific ML models, as in Yao et al. and Nguyen et al., further address local heterogeneity by tailoring predictions to individual clinic data, a strategy directly applicable to Sri Lankan fertility centers [3], [10].

Despite these advances, several gaps remain for a Sri Lanka–specific IVF prediction system. Key requirements include local calibration to account for regional variations in clinical practices and patient demographics, as Western models often underpredict in Asian contexts [4], [6]. Continuous learning pipelines are needed to adapt models to new outcomes and evolving practices, a gap noted in the limited governance details of existing studies [1], [3], [10]. Leakage control in multi-landmark models, as partially addressed in Grzegorczyk-Martin et al., is critical to avoid biased predictions, particularly when modeling outcomes conditional on reaching transfer [1]. data integration, such as aligning medical records with English clinical reports, is absent in the literature but essential for Sri Lanka's trilingual context [20]. Additionally, handling missing data and selection bias (e.g., dropout in multi-cycle predictions) is only explicitly addressed in a few studies, such as Ratna et al., which uses weighting and multiple imputation [5].

To address these gaps, this study proposes a multi-layered integration strategy combining fine-tuned logistic regression and ensemble ML models (e.g., Random Forest, XGBoost) with restricted cubic splines for nonlinear effects [2], [4], [20]. The system will incorporate dynamic multi-landmark updates inspired by Grzegorczyk-Martin et al. [1], cumulative live birth estimands from McLernon et al. [5], [6], [7], [8], [9], and center-specific calibration strategies from Nguyen et al. and Yao et al. [3], [10]. New modules will address Sri Lanka–specific needs, including local recalibration, continuous learning with drift monitoring. This approach aims to deliver a robust, explainable, and locally tailored IVF success prediction system for Sri Lankan couples, supporting personalized fertility treatment planning [12], [15], [18].

Figure 1- IVF Success Prediction System Architecture



1.2 Research Gap

The development of predictive models for In-Vitro Fertilization (IVF) success has advanced significantly, with multi-landmark modeling, machine learning (ML), and cumulative live birth rate (CLBR) prediction frameworks demonstrating promise in Western and Chinese populations [1]–[9], [12], [15], [18]. However, several critical gaps remain for creating a dependable, Sri Lanka-specific IVF success forecasting system that meets the needs of local couples and clinicians

First, local calibration for Sri Lankan populations is largely absent. Existing models, primarily trained on Western (e.g., UK, US) or Chinese datasets, fail to account for Sri Lanka's unique demographic, lifestyle, and clinical variations, such as regional dietary habits, stress factors, or genetic predispositions [4], [6], [7], [8], [9]. Studies like Cai et al. and Ratna et al. highlight systematic underprediction when applying Western models to Asian cohorts, underscoring the need for locally recalibrated models [4], [6]. Without Sri Lanka-specific data, predictions lack accuracy and relevance for local clinical decision-making.

Second, integration of lifestyle and environmental factors is insufficient. Most IVF prediction models focus on medical and hormonal markers (e.g., age, AMH, FSH, egg yield) but rarely incorporate lifestyle variables such as diet, physical activity, smoking/alcohol habits, or stress levels, which are critical in Sri Lanka's diverse socio-cultural context [1], [2], [3], [5]. For instance, Grzegorczyk-Martin et al. and Xia et al. use extensive medical feature sets but omit lifestyle factors, limiting their ability to capture holistic patient profiles [1], [2]. This gap restricts personalized predictions for Sri Lankan couples with varied lifestyles.

Third, dynamic multi-landmark modeling with continuous learning is underdeveloped. While Grzegorczyk-Martin et al. propose multi-stage landmark models (consultation, post-stimulation, post-lab, at transfer) [1], and McLernon et al. incorporate post-cycle revisions [5], [6], [7], [8], [9], these frameworks lack continuous learning mechanisms to adapt to new patient outcomes or evolving clinical practices. In Sri Lanka, where IVF practices and patient demographics may shift rapidly, static models risk obsolescence. Additionally, explicit governance for continuous model updates and drift monitoring is rarely addressed [1], [3], [10].

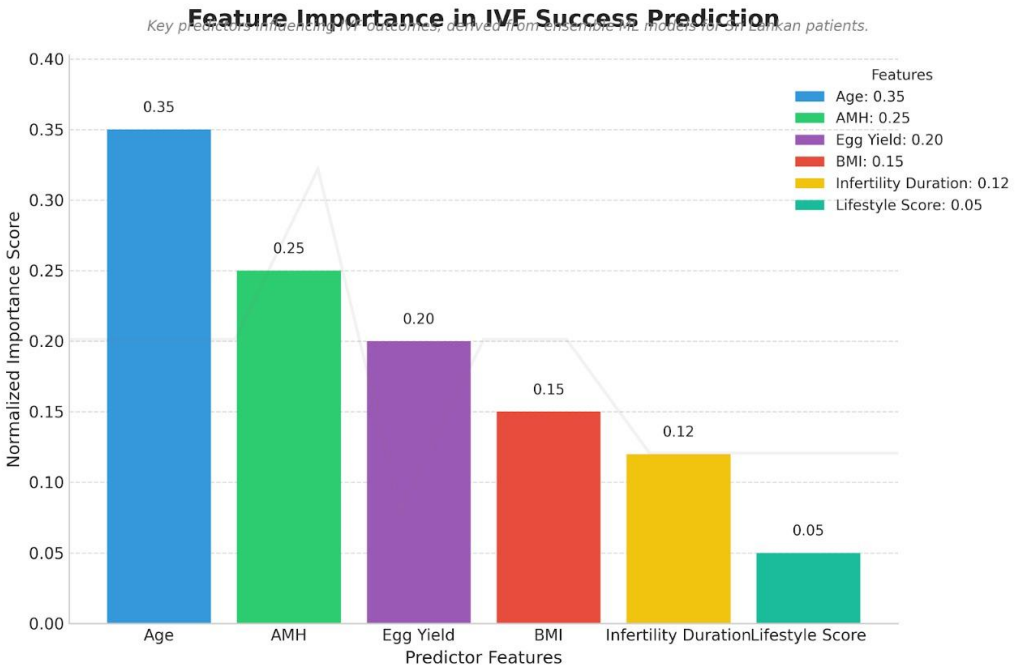
Fourth, data integration remains unaddressed**.** Sri Lankan medical records often contain inconsistencies in data formats, particularly due to manual entry or transcription errors. Existing models, developed for structured and standardized datasets (primarily English or Chinese), lack mechanisms for aligning heterogeneous inputs or correcting irregularities. This gap hinders the integration of comprehensive patient data critical for accurate predictions in Sri Lanka's healthcare context [2], [4], [20].

Fifth, explainability and local clinic-level calibration are insufficiently prioritized. While ensemble methods like Random Forest and XGBoost provide feature importance scores [3], [4], and restricted cubic splines offer interpretable nonlinear modeling [2], few studies focus on clinic-specific calibration or hierarchical modeling to account for variations across Sri Lankan fertility centers. Nguyen et al. and Yao et al. explore center-specific models but do not address Sri Lanka's unique clinic heterogeneity or provide patient-facing explainability tools [3], [10]. Transparent, locally calibrated predictions are essential for building trust and supporting shared decision-making.

Finally, handling of missing data and selection bias is inadequately addressed. Multi-cycle IVF predictions often suffer from informative censoring (e.g., patient dropout after failed cycles), yet only a few studies, such as Ratna et al., explicitly handle this through weighting or imputation [5]. Additionally, leakage control in multi-landmark models, as partially addressed in Grzegorczyk-Martin et al., is critical to avoid biased predictions but remains underexplored [1]. These issues are particularly relevant in Sri Lanka, where data collection may be incomplete due to resource constraints.

This research addresses these gaps by developing an AI-driven, dynamic IVF success forecasting system tailored for Sri Lanka. The system will: (a) recalibrate models using Sri Lankan patient data, (b) integrate medical, hormonal, and lifestyle factors, (c) implement dynamic multi-landmark modeling with continuous learning and drift monitoring, (d) provide explainable predictions with clinic-specific calibration, and (e) incorporate robust strategies for missing data and selection bias. By targeting these unmet needs, this work aims to deliver a personalized, adaptive, and locally relevant IVF success prediction tool for Sri Lankan couples and clinicians, advancing the application of AI in reproductive medicine.

Figure 2- Feature Importance in IVF Success Prediction



1.3 Research Problem

The increasing reliance on In-Vitro Fertilization (IVF) in Sri Lanka to address infertility highlights the urgent need for accurate, personalized, and locally relevant success prediction tools, given the procedure's high financial and emotional costs [7], [20]. Existing IVF prediction models, developed primarily for Western (e.g., UK, US) or Chinese populations, leverage advanced techniques such as multi-landmark modeling [1], machine learning (ML) algorithms like Random Forest and XGBoost [3], [4], [18], and cumulative live birth rate (CLBR) estimation over multiple cycles [5], [6], [8], [9]. However, these models are ill-suited for Sri Lanka due to unique demographic, lifestyle, clinical, and linguistic factors that influence IVF outcomes.

Specifically, existing IVF prediction systems fail to address several Sri Lanka–specific requirements:

1. Localized Prediction Models: Existing models, trained on non-Sri Lankan datasets, do not account for regional variations in patient demographics, genetic profiles, or clinical practices, leading to poor predictive accuracy [4], [6], [9]. For example, Cai et al. found systematic underprediction when applying Western models to Asian cohorts, underscoring the need for Sri Lanka-specific calibration [4].

2. Holistic Feature Integration: Most systems prioritize medical and hormonal markers (e.g., age, AMH, FSH, egg yield) but overlook lifestyle and environmental factors such as diet, physical activity, smoking, alcohol consumption, and stress, which are critical in Sri Lanka's diverse socio-cultural context [1], [2], [3]. This limits the ability to capture comprehensive patient profiles for personalized predictions.

3. Dynamic and Adaptive Modeling: While multi-landmark models (e.g., consultation, post-stimulation, post-lab, at transfer) exist [1], they lack continuous learning mechanisms to incorporate new patient outcomes or adapt to evolving IVF protocols in Sri Lanka's rapidly developing healthcare landscape [1], [3], [10]. Static models risk becoming outdated without drift monitoring or governance for updates.

4. Multilingual Data Processing: Sri Lankan patient records often combine Sinhala, Tamil, and English, particularly for medical histories and lifestyle data. Current systems, designed for monolingual datasets, lack robust methods to align multilingual inputs or handle inconsistencies from manual data entry or transcription errors [2], [4], [20], impeding comprehensive data integration.

5. Explainable and Clinic-Specific Predictions: Existing models provide limited patient-facing explainability and fail to account for heterogeneity across Sri Lankan fertility clinics [3], [4], [10]. Transparent, clinic-specific predictions with feature importance insights (e.g., impact of BMI or AMH) are essential for building trust and supporting shared decision-making in diverse clinical settings.

6. Robust Handling of Data Challenges: Issues such as missing data and selection bias (e.g., patient dropout after failed cycles) are inadequately addressed in most models, with only a few, like Ratna et al., using weighting or imputation strategies [5]. In Sri Lanka's resource-constrained settings, robust methods for handling incomplete data and informative censoring are critical for reliable predictions.

The research problem, therefore, is to develop an AI-driven, multilingual, and dynamic IVF success forecasting system tailored for Sri Lankan couples and clinicians. The system should:

1. Recalibrate ML models using Sri Lanka-specific patient data to ensure local relevance,
2. Integrate medical, hormonal, and lifestyle factors for holistic predictions,
3. Implement dynamic multi-landmark modeling with continuous learning and drift monitoring to adapt to new data and practices,
4. Align Sinhala, Tamil, and English data inputs while mitigating inconsistencies from manual entry or transcription,
5. Provide explainable predictions with feature importance dashboards and clinic-specific calibration to enhance trust and usability,

6. Address missing data and selection bias through robust statistical techniques, such as multiple imputation and weighting.

Addressing this problem will deliver accurate, personalized, and adaptive IVF success predictions for Sri Lankan couples, enabling informed decision-making, reducing emotional and financial burdens, and advancing the integration of AI in reproductive medicine within Sri Lanka's unique healthcare context.

2.0 Objectives

2.1 Main Objective

To develop a comprehensive, AI-driven forecasting platform for predicting IVF success probabilities tailored for Sri Lankan couples and clinicians, integrating medical, hormonal, and lifestyle data to provide dynamic, personalized, and explainable predictions. The platform will support informed decision-making, reduce emotional and financial burdens, and advance reproductive medicine in Sri Lanka's unique healthcare context.

2.2 Specific Objectives

1. Design and implement an end-to-end AI-based prediction pipeline tailored to Sri Lanka's IVF patient population and clinical practices.

2. Perform dynamic multi-landmark modeling to predict IVF success probabilities at key stages (consultation, post-stimulation, post-lab, at transfer), updating predictions as new data accrues.

3. Develop and compare multiple machine learning models (e.g., Logistic Regression, Random Forest, XGBoost) with restricted cubic splines to model nonlinear relationships and enhance predictive accuracy.

4. Align multilingual patient data (Sinhala, Tamil, English) from medical records, handling inconsistencies from manual entry or transcription errors to ensure robust data integration.

5. Provide explainable predictions through feature importance dashboards, highlighting key factors (e.g., age, BMI, AMH) influencing IVF outcomes for patients and clinicians.

6. Implement continuous learning mechanisms with drift monitoring to adapt the model to new patient outcomes and evolving clinical practices in Sri Lanka.

7. Address missing data and selection bias using robust statistical techniques, such as multiple

imputation and weighting, to ensure reliable predictions in resource-constrained settings.

8. Calibrate models for clinic-specific variations across Sri Lankan fertility centers, ensuring localized accuracy and applicability.

9. Evaluate model performance using standard metrics (accuracy, precision, recall, F1-score, ROC-AUC) on Sri Lankan IVF patient data and benchmark datasets.
10. Develop a user-friendly interface that delivers personalized IVF success probabilities, feature importance insights, and decision-support tools for couples and clinicians, preserving transparency and usability.

3.0 METHODOLOGY

3.1 Requirement Gathering

This phase combines a structured review of prior work on IVF outcome prediction, machine learning (ML) applications in reproductive medicine, and multi-landmark modeling with an analysis of Sri Lankan IVF patient data and clinical practices. The review identifies reusable components and limitations relevant to the local context, while the analysis of Sri Lankan IVF data surfaces domain-specific needs,integration of lifestyle factors, clinic-specific calibration, continuous learning, and robust handling of missing data and selection bias. The outputs are a requirements checklist, an initial data schema for patient profiles, and a baseline selection of ML models for comparison.

3.1.1 Past Research Analysis

Existing literature demonstrates effective IVF success prediction through multi-landmark models [1], ensemble ML methods like Random Forest and XGBoost [3], [4], [18], and cumulative live birth rate (CLBR) frameworks [5], [6], [7], [8], [9]. Studies such as Grzegorczyk-Martin et al. showcase dynamic modeling across IVF stages (consultation, post-stimulation, post-lab, at transfer) [1], while Xia et al. use restricted cubic splines for nonlinear effects of age and AMH [2]. However, gaps remain for Sri Lanka: (i) local calibration for Sri Lankan patient demographics and clinical practices, (ii) integration of lifestyle factors (e.g., diet, physical activity, stress), (iii) continuous learning with drift monitoring, (iv) multilingual data alignment for Sinhala, Tamil, and English records, (v) clinic-specific calibration, and (vi) robust handling of missing data and selection bias. Accordingly, prior methods are adopted where suitable (e.g., multi-landmark frameworks, ensemble ML with restricted cubic splines, feature importance analysis) and extended to meet Sri Lankan requirements. For example, logistic regression and XGBoost models will be recalibrated using local data, multilingual alignment techniques will be developed for data integration, and continuous learning pipelines will be implemented to adapt to new outcomes. This analysis directly informs the technical requirements, data specification, and evaluation plan for the proposed system.

3.2 Feasibility Study

3.2.1 Technical Feasibility

3.2.1.1 Knowledge of Technologies

1. Data Preprocessing: Enables extraction and cleaning of patient data from structured (e.g., medical records) and unstructured (e.g., clinical notes) sources, utilizing techniques such as data normalization and handling missing values through imputation and weighting.
2. Machine Learning Modeling: Applies algorithms including Logistic Regression, Random Forest, and XGBoost with restricted cubic splines to predict IVF success probabilities, ensuring nonlinear relationships (e.g., age, AMH) are captured effectively.
3. Dynamic Multi-Landmark Prediction: Implements stage-specific models fine-tuned to update success probabilities at key IVF stages (consultation, post-stimulation, post-lab, pre-transfer), leveraging time-series data integration.
4. Feature Importance Analysis: Extracts and ranks key predictors (e.g., age, BMI, egg yield) using ensemble methods, providing explainable insights through feature importance dashboards for clinicians and patients.
5. Continuous Learning: Incorporates drift monitoring and model retraining techniques to adapt predictions to new patient outcomes and evolving clinical practices in Sri Lanka.
6. Mobile App Integration: Utilizes React Native to develop a cross-platform interface, integrating machine learning models (e.g., TensorFlow.js) for on-device inference and interactive visualizations (e.g., React Native Chart Kit) for success probabilities and dashboards.
7. Clinic-Specific Calibration: Applies hierarchical modeling or transfer learning to tailor predictions to individual Sri Lankan fertility centers, ensuring localized accuracy and relevance.

3.2.1.2Knowledge on Tools

1. Data Preprocessing: Pandas, NumPy, Scikit-learn
2. Machine Learning Libraries: Scikit-learn, XGBoost, SciPy
3. Deep Learning: PyTorch, TensorFlow
4. Mobile App Development: React Native, Firebase, Android Studio, Xcode
5. Storage & Compute: MongoDB Atlas, Google Colab, AWS, Google Cloud
6. Evaluation & Visualization: Matplotlib, Seaborn, Scikit-learn
7. Annotation Tools: Label Studio, Prodigy
8. Project Management: Trello, Jira, Notion, GitHub

3.3 System Analysis

3.3.1 Software Solution Approach

The proposed solution adopts a multi-stage pipeline for predicting IVF success probabilities tailored for Sri Lankan couples and clinicians, delivered as a cross-platform mobile application developed using Flutter for the frontend and Node.js with Express and MongoDB for the backend. The system integrates medical, hormonal, and lifestyle data from open-source datasets and locally collected data from doctors to provide dynamic, personalized, and explainable predictions. Figure X illustrates the overall workflow of the solution.

1. Data Collection: Aggregate IVF patient data from open-source repositories (e.g., anonymized fertility registries, public medical research datasets like SART CORS) and locally collected data from Sri Lankan doctors, including medical records (e.g., AMH, FSH, egg yield), hormonal profiles (e.g., LH, Estradiol), and lifestyle questionnaires (e.g., BMI, diet, stress levels). Data will cover multilingual (Sinhala, Tamil, English) sources to ensure comprehensive representation of Sri Lankan patient profiles.

2. Data Ingestion: Upload data in structured formats (e.g., JSON, CSV) and unstructured formats (e.g., text-based medical notes) via the Flutter mobile app or clinic web portals, supporting batch processing for clinic datasets and real-time input for individual patient profiles, with data stored in MongoDB for flexible schema handling.

3. Data Preprocessing : Apply text preprocessing using Node.js libraries (e.g., *Natural*, *Compromise*) to clean and normalize medical records. This includes removing noise (extra spaces, punctuation errors), correcting misspellings from manual entry, and standardizing medical terminology.
4. Feature Extraction and Integration: Extract and encode features from both open-source and doctor-provided datasets, including medical (e.g., ovarian reserve, sperm motility), hormonal (e.g., Estradiol levels), and lifestyle factors (e.g., physical activity, smoking habits), using Node.js scripts to process data and store feature vectors in MongoDB for model input.

5. Dynamic Multi-Landmark Modeling: Implement machine learning models (Logistic Regression, Random Forest, XGBoost) with restricted cubic splines using Scikit-learn or TensorFlow.js, hosted on the Node.js/Express backend, to predict IVF success probabilities

at key stages (consultation, post-stimulation, post-lab, at transfer), updating predictions dynamically as new data accrues during the IVF cycle.

6. Explainability and Visualization: Generate feature importance dashboards using SHAP and LIME, rendered via Flutter's charting libraries (e.g., fl_chart) to highlight key predictors (e.g., age, AMH, BMI), and create interactive visualizations to display success probabilities and model insights for patients and clinicians on the mobile app.

7. Continuous Learning and Drift Monitoring: Deploy a continuous learning pipeline using Node.js with MLflow to update models with new patient outcomes from open-source and doctor-collected data, stored in MongoDB, incorporating drift monitoring to adapt to changes in clinical practices or patient demographics in Sri Lanka.
8. Clinic-Specific Calibration: Fine-tune models for individual Sri Lankan fertility clinics using MongoDB queries to segment clinic-specific data from locally collected records, ensuring localized prediction accuracy within the Node.js/Express backend.
9. Evaluation and Validation: Assess system performance using a combined dataset of open-source and Sri Lankan doctor-collected IVF patient data, supplemented by benchmark datasets, applying metrics (accuracy, precision, recall, F1-score, ROC-AUC) computed via Scikit-learn or custom Node.js scripts, with results logged in MongoDB for iterative improvement.

10. Mobile App Deployment: Develop a cross-platform mobile app using Flutter to deliver personalized IVF success probabilities, feature importance insights, and decision-support tools, with a Node.js/Express backend and MongoDB for real-time data processing and storage, ensuring a seamless and user-friendly experience for patients and clinicians.

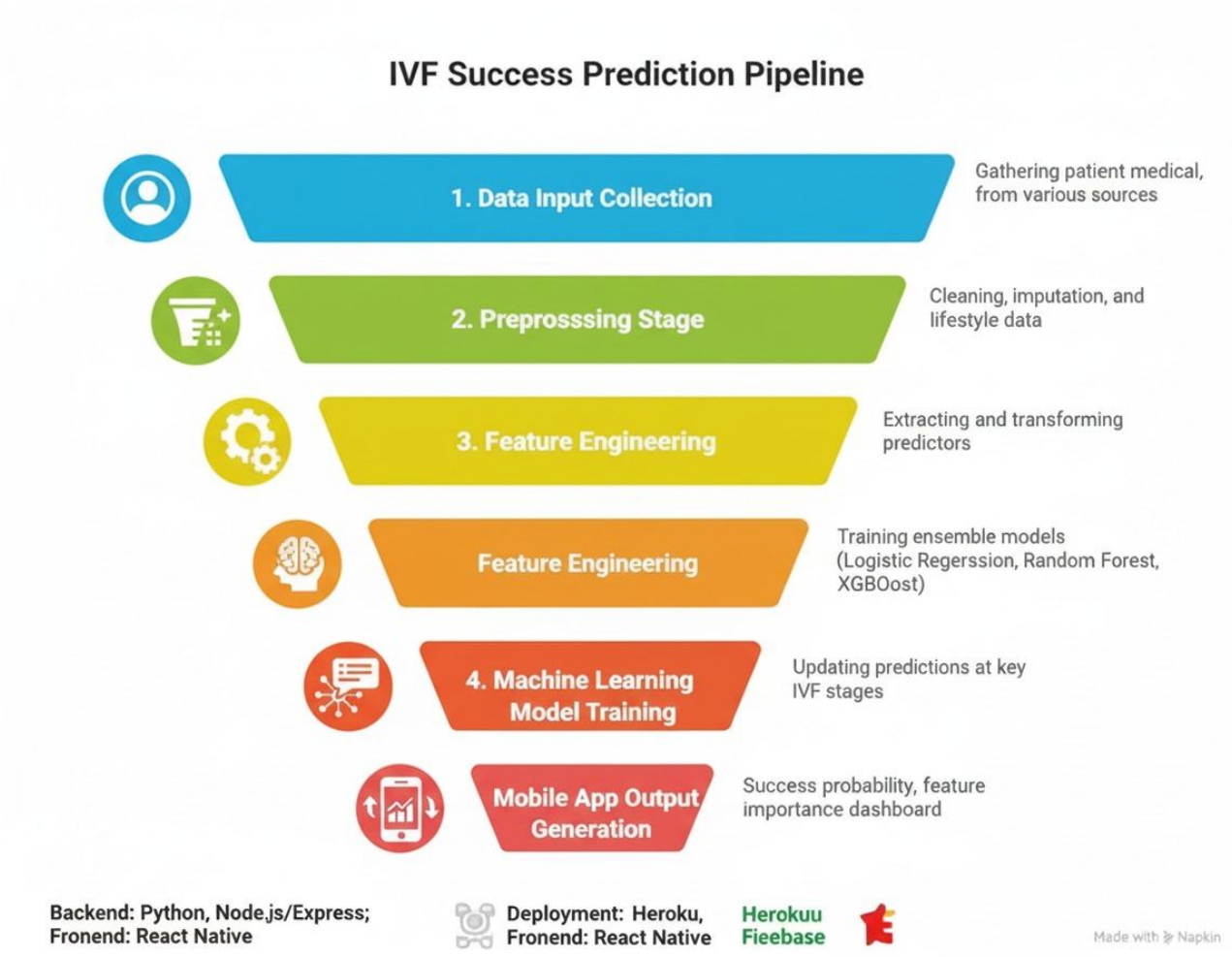


Figure 3: System Architectural diagram

3.3.2 Tools & Technology

1. Programming: JavaScript/TypeScript (React Native for mobile app frontend, Node.js with Express for backend), Python (machine learning model development and data preprocessing).

2. Machine Learning & Data Processing: Scikit-learn (Logistic Regression, Random Forest), XGBoost (gradient boosting), TensorFlow.js (advanced neural networks optimized for React Native integration), SciPy (restricted cubic splines for nonlinear modeling), Pandas and NumPy (data cleaning, feature engineering, and handling missing data via imputation/weighting).
3. Databases:
    - MongoDB Atlas (free tier for flexible storage of structured patient data, feature vectors, and model predictions).
    - SQLite (lightweight, local storage for mobile app caching to support offline functionality).
4. Mobile Front End: React Native (cross-platform Android/iOS app development), React Native Chart Kit (interactive visualizations for feature importance and success probabilities), NativeBase or React Native Paper (for consistent, user-friendly UI styling).
5. Design/Diagrams: Figma (UI/UX design for mobile app mockups), draw.io (system architecture and workflow diagrams).
6. IDE: Visual Studio Code (JavaScript/TypeScript, Python development), Android Studio (React Native debugging, emulation, and Android testing), Xcode (iOS testing, if needed).
7. Version Control System (VCS): GitHub (code and data schema versioning, public repositories for cost efficiency).
8. Collaboration: WhatsApp (team communication, widely used in Sri Lanka), Google Drive (document sharing and storage).
9. Testing: Jest (JavaScript/TypeScript testing for React Native and backend), pytest (Python ML model testing), Detox (React Native end-to-end testing), Postman (API endpoint testing).
10. Deployment: Firebase (free tier for React Native app hosting, real-time database, and authentication), Node.js/Express on Heroku (backend deployment, free tier), Docker (optional, for containerizing backend services).
11. Code Quality: ESLint (JavaScript/TypeScript linting), Flake8 (Python linting), Prettier (code formatting for React Native).
12. Project Management: Trello (collaborative task tracking and planning), GitHub Projects

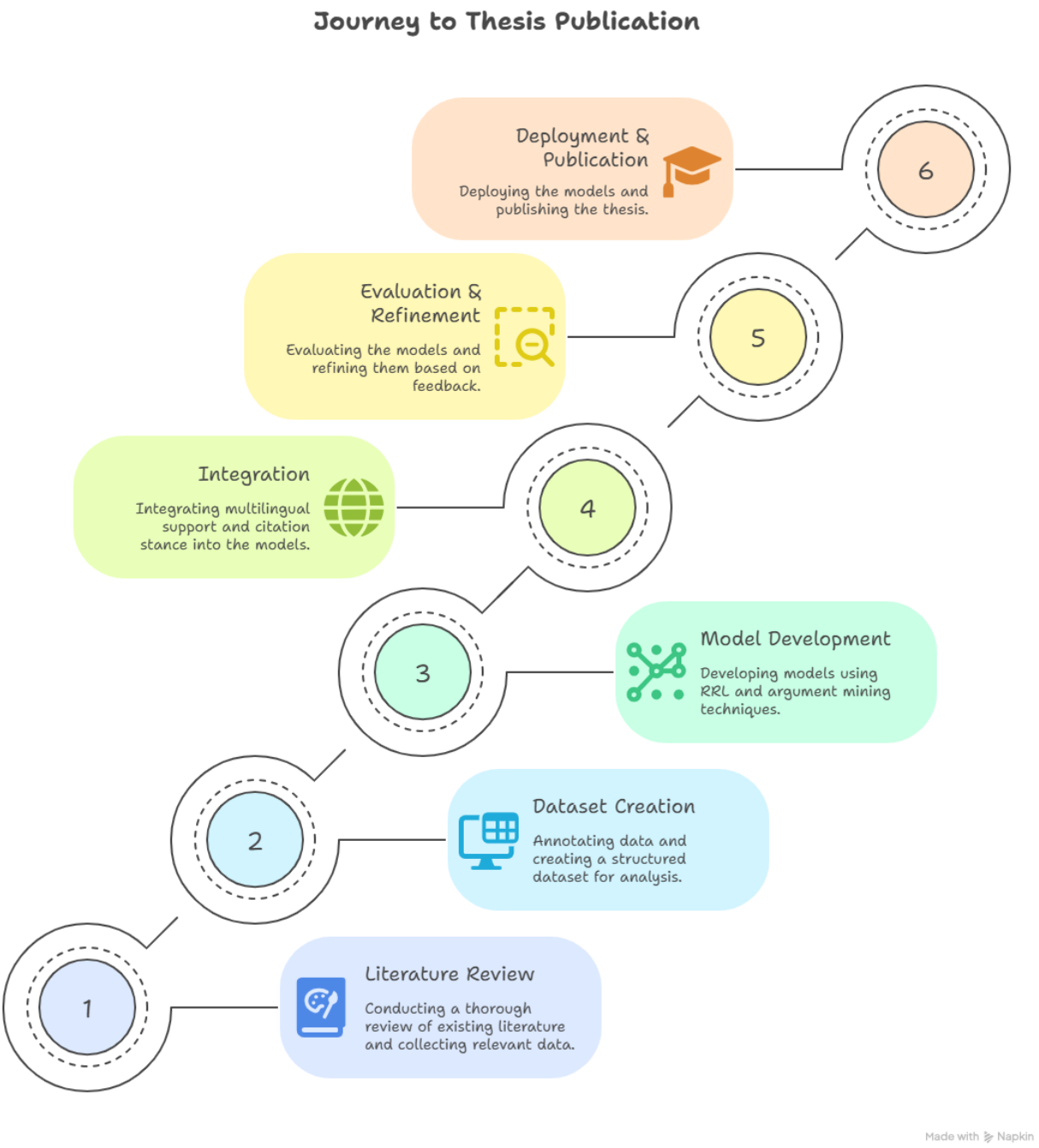


Figure 4: Work breakdown Structure

3.4 Project Requirements

3.4.1 Functional Requirements

3.4 Project Requirements

3.4.1 Functional Requirements

Requirement	Description
Data Ingestion	Ingest IVF patient data from open-source repositories and locally collected data from Sri Lankan doctors in structured (e.g., JSON, CSV) and unstructured (e.g., text-based medical notes) formats via the Flutter mobile app or clinic web portals.
Data Preprocessing	medical records, aligning data from diverse sources and handling inconsistencies from manual entry or transcription errors using Node.js libraries
Feature Extraction	Extract and encode medical , hormonal, and lifestyle factors from both open-source and doctor-collected datasets, storing feature vectors in MongoDB for model input.
Dynamic Prediction Modeling	Implement machine learning models with restricted cubic splines to predict IVF success probabilities at key stages (consultation, post-stimulation, post-lab, at transfer), updating predictions dynamically as new data accrues.
Explainability and Visualization	Generate feature importance dashboards using SHAP and LIME, rendered via Flutter's charting libraries (e.g., fl_chart), to highlight key predictors (e.g., age, AMH, BMI) and display interactive success probability visualizations for patients and clinicians.
Continuous Learning	Deploy a continuous learning pipeline using Node.js with MLflow to update models with new patient outcomes from MongoDB, incorporating drift monitoring to adapt to changes in clinical practices or patient demographics.
Clinic-Specific Calibration	Fine-tune models for individual Sri Lankan fertility clinics using MongoDB queries to segment clinic-specific data, ensuring localized prediction accuracy.
Structured Outputs	Generate structured outputs (e.g., JSON) containing success probabilities, feature importance scores, and clinic-specific insights for search, analysis, and integration with clinic systems.

3.4.2 Non-Functional Requirements

Requirement	Description
Usability	Provide an intuitive Flutter-based mobile app interface for patients and clinicians, with clear visualizations and decision-support tools, accessible to users with varying technical expertise.
Reliability	Ensure consistent prediction results across diverse data formats (structured and unstructured) and noisy inputs (e.g., incomplete medical records), with robust error handling in the Node.js/Express backend.
Availability	Deploy the system on cloud platforms (e.g., Firebase, AWS Amplify, Heroku) for 24/7 access, ensuring minimal downtime for users in Sri Lanka and beyond.
Accuracy	Achieve high accuracy, precision, recall, F1-score, and ROC-AUC for prediction tasks, validated using combined open-source and Sri Lankan doctor-collected datasets.
Performance	Support efficient batch processing of clinic datasets and real-time predictions for individual patients, with acceptable response times (e.g., <2 seconds for single predictions) on the mobile app.
Scalability	Design the system to handle increasing volumes of patient data, additional clinics, and potential expansion to other regions, leveraging MongoDB's flexible schema and Node.js's asynchronous processing.
Security	Protect sensitive patient data with encryption , comply with medical data privacy regulations, and implement secure user authentication in the app.

3.5 Testing

Testing is a pivotal phase to ensure the robustness, accuracy, and usability of the proposed IVF success forecasting system tailored for Sri Lankan couples and clinicians. The testing strategy will be conducted in multiple phases to validate the system's functionality, performance, and clinical relevance:

1. Unit Testing: Individual components, including data preprocessing, , feature extraction (e.g., AMH, BMI, lifestyle factors), and machine learning models (Logistic Regression, Random Forest, XGBoost), will be tested with controlled inputs. For instance, model predictions will be checked against known IVF outcomes from benchmark datasets.
2. System Testing: The end-to-end pipeline (from data ingestion to prediction output and visualization) will be tested using a combined dataset of open-source IVF data (e.g., SART CORS) and locally collected data from Sri Lankan doctors to ensure seamless integration across Flutter frontend, Node.js/Express backend, and MongoDB.

3. Expert Validation: Fertility specialists, clinicians, and medical researchers will evaluate prediction outputs, feature importance dashboards (generated via SHAP/LIME), and visualizations (via fl_chart) to confirm clinical accuracy, relevance, and interpretability for Sri Lankan patient profiles.

4. Performance Testing: Assess system performance under varying loads, such as batch processing 100+ patient records and real-time predictions for individual profiles, ensuring response times remain below 2 seconds per prediction and efficient MongoDB query performance.

5. Acceptance Testing: Deploy the prototype in a cloud environment (e.g., Firebase, AWS Amplify) for testing by selected users, including patients, clinicians, and medical students. User feedback on the Flutter app's usability and decision-support tools will drive iterative refinements.

6. Testing will integrate manual validation by medical experts with automated evaluation metrics (accuracy, precision, recall, F1-score, ROC-AUC) to assess prediction

accuracy, feature importance reliability, and data alignment, ensuring robust performance across diverse data inputs.

3.6 Timeline

The proposed timeline for the project is as follows.

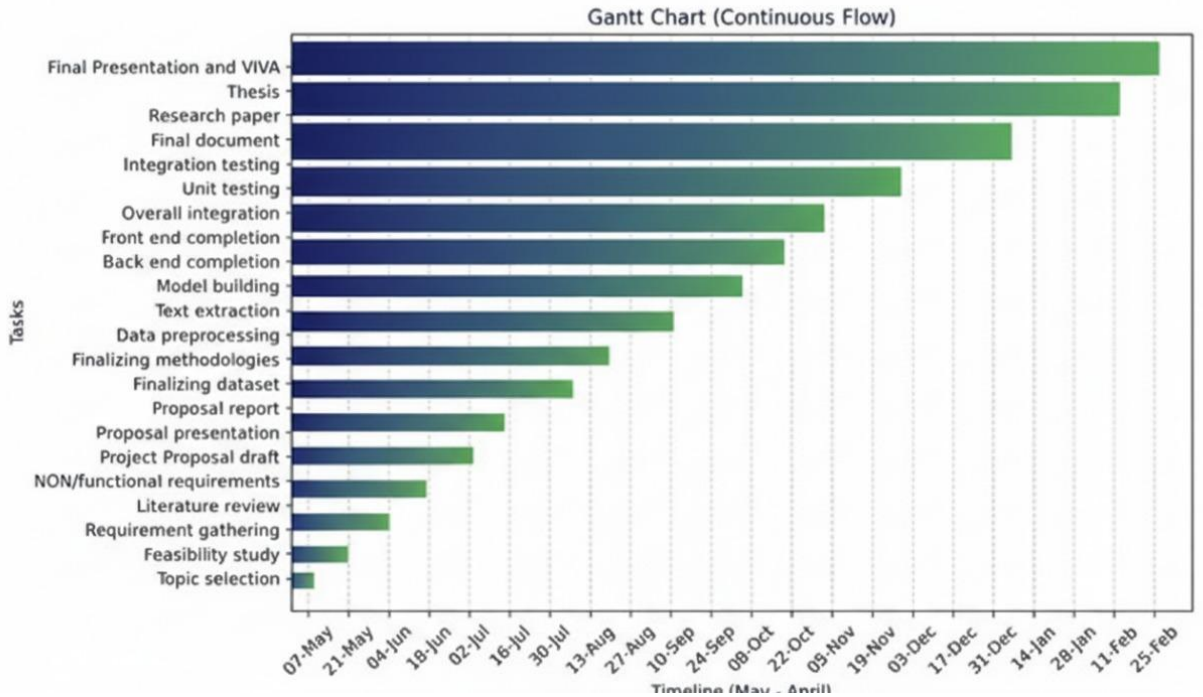


Figure 5:Gantt Chart for Project Schedule Management

3.7 Risk Management Plan

The risk management plan outlines strategies to address potential challenges in developing the IVF success forecasting system, ensuring the development team completes the project without significant interruptions. The table below details typical risks, their probability, impact, priority, ownership, mitigation strategies, and contingency plans specific to the project's context.

Table 1-: Risk Management Plan

Risk	Description	Probability	Impact	Priority	Owner	Mitigation Strategy	Contingency Plan
Financial	Exceeding budget due to unexpected costs (e.g., cloud hosting, data acquisition).	Medium	High	High	Project Manager	Track expenses using Jira, set strict budget limits, and prioritize cost-effective tools (e.g., Firebase free tier).	Secure a reserve fund and reduce non-critical expenses .
Technical	Bugs or failures in core components	Medium	High	High	Technical Lead	Perform thorough unit, integration, and performance testing with pytest, Jest, and Flutter Test, alongside regular code reviews.	Revert to a stable version and allocate extra developer resources for rapid fixes.
Operational	Loss of a key team member during a critical phase.	Low	High	Medium	Team Lead	Cross-train team members on essential tasks and document processes in GitHub for continuity.	Recruit backup freelancers or part-time developers for quick onboarding.
Data Quality	Incomplete or noisy data from open-source datasets or doctor-collected records, reducing model accuracy	Low	Medium	Low	Data Scientist	Validate data during preprocessing, apply imputation techniques (e.g., multiple imputation), and use benchmark datasets for robustness.	Expand data collection by partnering with additional clinics or sourcing more open-source data. .

3.8 Communication Management Plan

This plan defines who needs what information, how it's shared, when it's shared, and who

is responsible—so the supervisor, co-supervisor, team, and external advisors can make timely decisions and keep the project on track.

3.8.1 Communication Objectives:

Proactive communication will be:

- Targeted – matched to the stakeholder's role and decisions needed.
- Adequate – correct detail and format; no missing context.
- Sufficient – all necessary attachments, links, and data included.
- Concise – brief, action-oriented, avoids repetition.
- Timely – sent early enough to act; uses agreed cadences and SLAs.

3.8.2 Communication Media:

- Primary: Microsoft Teams (meetings/chat), Email (formal notices/approvals).
- Work tracking: Click Up/Trello (tasks), GitHub (code, Issues, PRs, Wiki).
- Quick updates: WhatsApp (urgent coordination only).
- Documents: Google Drive/OneDrive folders with version history.
- Meeting minutes: shared Doc (dated; action items with owners & due dates).

Table 2- Communication Management Plan

Audience	Content	Format/Channel	Frequency	Owner	Success Criteria
Supervisor & Co-supervisor	Progress, risks, decisions needed, upcoming milestones	Slide deck + email; MS Teams meeting	Bi-weekly	Project Lead	Actions agreed, shared ≤24h

Project Team	Tasks, blockers, code merges, data updates	Click Up/Trello updates; GitHub Issues/PRs; Teams chat	Weekly + as needed	Module Owners	No overdue critical tasks; blockers resolved within SLA
Legal Experts	Samples of holdings, statute-outcome links, citation stance errors	Brief memo + annotated examples; Teams call	Monthly / per batch	IE Lead	≥90% agreement on sampled items

4.0 COMMERCIALIZATION

The commercialization strategy targets fertility clinics, reproductive medicine specialists, and patients seeking IVF treatment as primary users, with secondary users including medical researchers, healthcare policymakers, and fertility app developers. The system differentiates itself through personalized, explainable IVF success predictions, clinic-specific calibration tailored to Sri Lanka. Offered as a Software-as-a-Service (SaaS) model, it provides basic, professional, and enterprise tiers for individual patients, clinicians, and clinics, respectively, alongside API access for integration with healthcare systems. Go-to-market efforts include pilots with Sri Lankan fertility clinics, partnerships with medical associations, and outreach to universities for research adoption. Reliable data sources will be drawn from open-source fertility registries (e.g., SART CORS) and locally collected data from doctors, with MongoDB, Node.js, and cloud providers (e.g., Firebase, AWS Amplify) supporting the technology stack. Compliance is ensured through adherence to Sri Lankan medical data privacy regulations, robust encryption, and human-in-the-loop model validation. Key risks such as data quality issues, regulatory changes, adoption barriers, and cost overruns will be mitigated with rigorous data preprocessing, legal consultations, user training programs, and optimized cloud resource management.

5.0 BUDGET

Focus on essential, low-cost items. Free/academic tiers are used where possible; a small cloud buffer is included only for occasional paid usage.

Table 3-Cost management plan

Category	**Cost (LKR)**	**Justification**
Mobile App Hosting & Backend	4,000	Free tiers for backend hosting (e.g., Firebase, Heroku) for API and database; cost covers annual domain for API endpoints or minimal paid hosting if free tier limits are exceeded.
Development tools & repositories	0	Free tools: VS Code, GitHub , Android Studio, Xcode (for iOS testing), and React Native for cross-platform development. MongoDB Atlas for data storage
Cloud services (compute/storage)	7,000	Minimal paid usage beyond free tiers (e.g., AWS, Google Cloud) for ML model inference (e.g., Random Forest, XGBoost) and storage for anonymized patient data or model snapshots.
App Store Fees	3,500	One-time Google Play Store fee -Minimal costs for app updates
Miscellaneous & contingency	3,500	Incidental expenses (e.g., local travel for clinic visits, testing devices, minor software licenses); ~20% buffer for unforeseen costs like additional cloud usage or app testing.
Total	18,000	—

6.0 SUMMARY

The rising demand for In Vitro Fertilization (IVF) in Sri Lanka, coupled with limited access to advanced fertility treatments and high emotional and financial costs, necessitates accurate, personalized, and locally relevant success prediction tools. Existing IVF prediction models, primarily developed for Western or Chinese populations, fail to account for Sri Lanka's unique demographic, lifestyle, and clinical factors, resulting in underprediction and limited applicability. This project proposes an AI-driven mobile app for IVF success prediction that is context-centric: it integrates medical, hormonal, and lifestyle data, provides dynamic multi-landmark predictions (consultation, post-stimulation, post-lab, pre-transfer), and ensures explainability through feature importance dashboards. The app aims to deliver verifiable, clinic-calibrated predictions suitable for personalized treatment planning and shared decision-making.

The methodology combines preprocessing of local IVF patient data with transformer-based machine learning models (e.g., Logistic Regression, Random Forest, XGBoost) enhanced by restricted cubic splines for nonlinear effects, dynamic multi-landmark modeling, and continuous learning with drift monitoring. Robust statistical techniques, such as multiple imputation and weighting, address missing data and selection bias. Models will be recalibrated using Sri Lankan IVF patient data and evaluated with standard metrics (accuracy, precision, recall, F1-score, ROC-AUC) and clinical validation. Expected outcomes include: (i) a reproducible prediction pipeline and gold-standard Sri Lankan IVF dataset, (ii) practitioner- and patient-oriented features such as dynamic success probabilities, feature importance dashboards, clinic-specific calibration, and decision-support tools, and (iii) a user-friendly mobile app interface that preserves data privacy while delivering structured, explainable predictions. These results aim to reduce emotional and financial burdens, enhance trust in predictions, and strengthen personalized fertility treatment planning for Sri Lankan couples and clinicians.

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